Experiment 16: Birds Fly Knowledge Base

Aim:

To implement a knowledge base with inheritance to determine whether an animal can fly based on its characteristics.

Algorithm:

1. Define class hierarchy: Bird -> Penguin, Ostrich, Sparrow.
2. Default rule: Birds can fly.
3. Exceptions: Penguins and Ostriches cannot fly.
4. Use inheritance with overriding for exceptions.

Code:

```
class Bird:
    def can_fly(self): return True

class Sparrow(Bird):
    pass

class Penguin(Bird):
    def can_fly(self): return False

class Ostrich(Bird):
    def can_fly(self): return False

animals = [Sparrow(), Penguin(), Ostrich()]
names = ['Sparrow', 'Penguin', 'Ostrich']
for name, animal in zip(names, animals):
    print(f"{name} can fly? {animal.can_fly()}")
```

Sample Output

```
Sparrow can fly? True
Penguin can fly? False
Ostrich can fly? False
```

Result:

The knowledge base correctly applied inheritance and exceptions to determine flying ability of different birds.